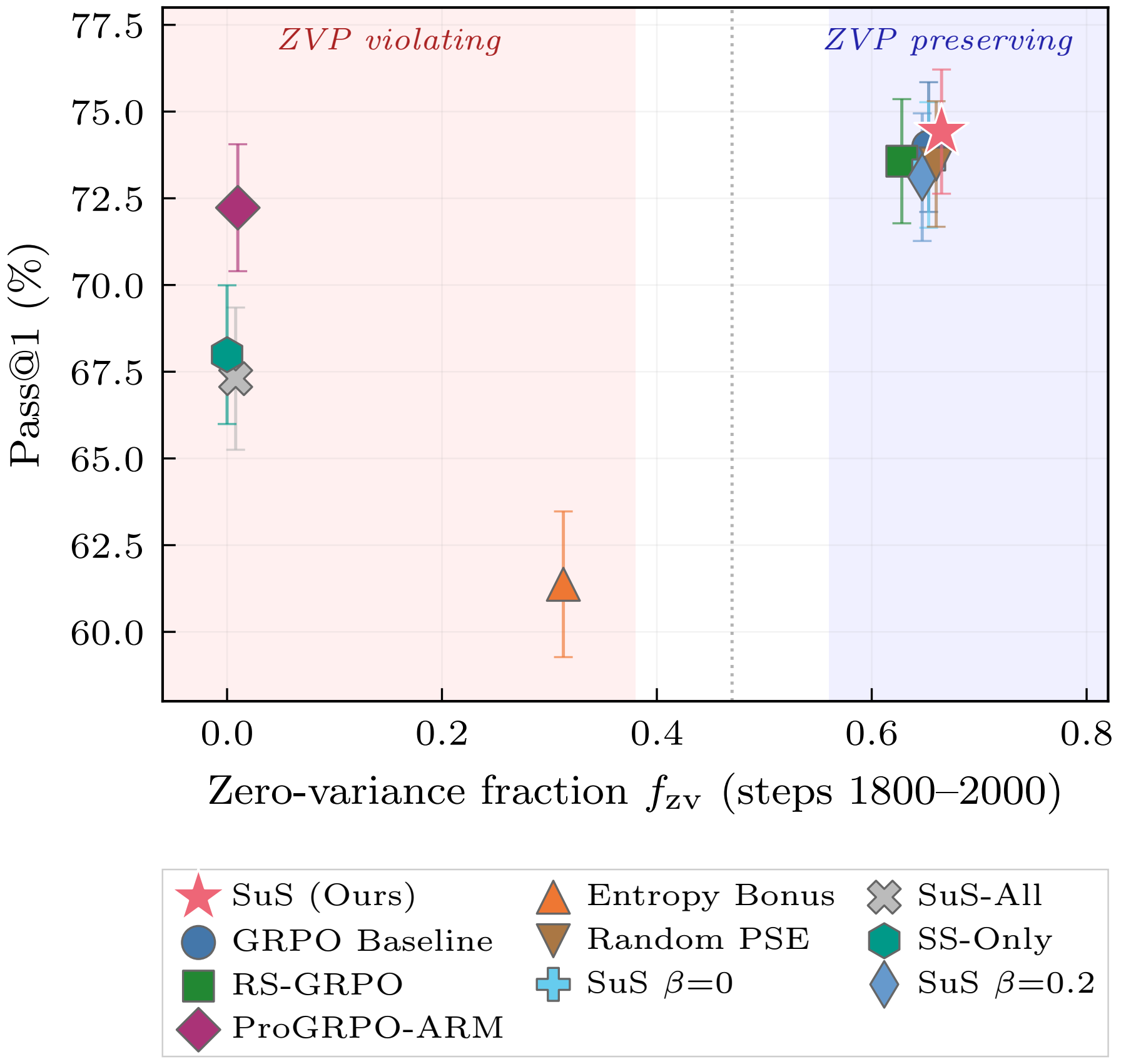
ZVP violating
ZVP preserving
Pass@1 (%)
77.5
75.0
72.5
70.0
67.5
65.0
62.5
60.0
0.0
0.2
0.4
0.6
0.8
Zero-variance fraction f_{zv} (steps 1800–2000)
SuS (Ours)
GRPO Baseline
RS-GRPO
ProGRPO-ARM
Entropy Bonus
Random PSE
SuS β=0
SuS-All
SS-Only
SuS β=0.2